

Q1. Hospital Patient Segmentation.

Given SSE values:

K	SSE
1	600
2	350
3	220
4	200
5	190

The elbow method is used to determine the optimal number of clusters by observing where the decrease in SSE becomes significantly smaller.

- from $k=1$ to $k=2$, SSE decreases from 600 to 350
- from $k=2$ to $k=3$, SSE decreases from 350 to 220
- from $k=3$ to $k=4$, SSE decreases only from 220 to 200
- from $k=4$ to $k=5$, SSE decreases only from 200 to 190

Therefore, the major reduction in SSE occurs up to $k=3$, after which the improvement becomes small.

Optimal number of clusters = 3

The patients can therefore be segmented into three groups based on medical expenses and health risk scores.

Effect on healthcare decision-making:

- Low-risk / low-expense patients can be managed through routine check-ups.
- Moderate-risk patients can receive regular monitoring and preventive care.
- High-risk / high-expense patients can be given priority treatment, closer monitoring and specialized healthcare services.

Thus, selecting 3 clusters provides meaningful patient segmentation while avoiding unnecessary complexity. It helps the healthcare organization allocate resources efficiently and design personalized treatment strategies:

2. E-commerce Order Analysis

Hierarchical clustering groups observation by calculating their similarity or distance and progressively joining the most similar observations. A dendrogram represents this process.

From the dataset:

- Electronics: $O1 = 55,000$ and $O3 = 57,000 \rightarrow$ very similar order values, so they are likely to form one cluster.
- Clothing: $O2 = 25,000$ and $O5 = 27,000 \rightarrow$ very similar order values, so they are likely to form another cluster.

* furniture: 04 = 30,000 → relatively different from the other categories and may form a separate cluster initially

By cutting the dendrogram at a suitable height, an appropriate solution is:

optimal number of clusters = 3

The clusters can be interpreted as:

cluster	Orders	Interpretation
cluster 1	01, 03	High-value Electronics orders.
cluster 2	02, 05	Medium-value clothing orders.
cluster 3	04	furniture order segment.

Significance of the clusters

These clusters help the retailer understand different purchasing patterns:

- Electronics cluster.
- clothing cluster.
- furniture cluster.

Thus, hierarchical clustering helps the retailer identify distinct customer order segments, improve targeted marketing, optimize promotions and better

1. The first part of the report

2. The second part of the report

3. The third part of the report

4. The fourth part of the report

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6. The sixth part of the report

7. The seventh part of the report

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17. The seventeenth part of the report

...and ...

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Q2

Although the user has less time on the application and log in less frequently, the user has the highest purchase value. The company can focus on:

- personalized purchase offers.
- premium products recommendations
- retention campaigns.
- Notification encouraging more frequent app usage.

Q4. Agricultural Crop Monitoring

Ans) Before normalization, rainfall has a much larger numerical range so it can dominate distance calculations and affect clustering. After normalization, All features are converted to a comparable scale. Therefore, rainfall, soil fertility and crop yield contribute more equally to cluster formation.

for example:

- F1 and F4 have high soil fertility and high crop yield
- F2 and F5 have soil fertility and lower crop yield.
- F3 has intermediate values.

Effects of normalization:

1. prevents large-scale features from dominating
2. Gives equal importance to different variables.
3. Produces more meaningful clusters
4. Improves the reliability of agricultural analysis

Q8) Employee performance evaluation

Ans) The given silhouette scores are:

• $k=2 \rightarrow 0.48$.

• $k=3 \rightarrow 0.68$

• $k=4 \rightarrow 0.52$

The best model is $k=3$, because it has the highest silhouette score of 0.68. A higher silhouette score indicates better separation between clusters and greater similarity among employees within the same cluster.

THE FIRST OF THE MONTH

THE SECOND OF THE MONTH

THE THIRD OF THE MONTH

THE FOURTH OF THE MONTH

THE FIFTH OF THE MONTH

THE SIXTH OF THE MONTH

THE SEVENTH OF THE MONTH

THE EIGHTH OF THE MONTH

THE NINTH OF THE MONTH